

مساحة الدائرة

Q4

$E = 2 \text{ V/m}$ at 10 m from isotropic source.

$$A = 4\pi \times 10^2 = 400\pi$$

b) Intensity is: $I = \frac{E^2}{2\mu_0 c} = \frac{2^2}{240\pi} =$

c) Power is: $P = I \times A = \left(\frac{2^2}{240\pi} \times 400\pi \right) = \underline{6.7 \text{ W}}$

بسم الله الرحمن الرحيم

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Q₄

$B_{\text{max}} = 2 \mu\text{T}$ at 10 m from isotropic source. The power is:

$$P = \left(\frac{E^2}{2\mu_0 c} \right) \cdot A$$

- 6) An electron is needed to be accelerated from rest to 40% of speed of light. The potential difference is:

$$v = \frac{40}{100} \times 3 \times 10^8 = 1.2 \times 10^8 \text{ m/s}$$

$$q(V_p) = \frac{1}{2} m v^2$$

$$V_p = \frac{m v^2}{2q} = \frac{9.11 \times 10^{-31} \times (1.2 \times 10^8)^2}{2 \times 1.6 \times 10^{-19}} = 40995 \text{ V}$$

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A

Proton is needed to be accelerated from rest to 50% of speed of light. The potential difference is:

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$$E_y = 118 \sin\left(\frac{k}{10^7}x - \omega t\right)$$

a) Magnetic field is : $B_z = \frac{E}{c} = \frac{118}{3 \times 10^8} = 39.3 \times 10^{-8} \text{ T at } z$

$k \rightarrow$ wave number (m^{-1})

$$\lambda = \frac{2\pi}{k}$$

$$f = \frac{c}{\lambda}$$

b) Frequency is : $\lambda = \frac{2\pi}{k} = \frac{2\pi}{10^7} = 2\pi \times 10^{-7} \text{ m}$

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{2\pi \times 10^{-7}} = 0.47 \times 10^{15} \text{ Hz}$$

مسألة رقم 7

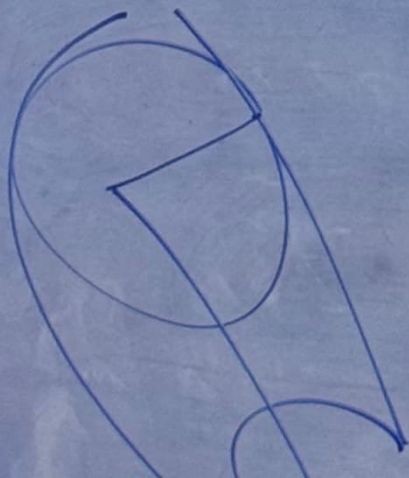
(7)

$$\psi = 118 \sin \pi(2y/3 - \omega t)$$

$k \rightarrow$ wave number (m^{-1})

$$\lambda = \frac{2\pi}{k}$$

$$f = \frac{c}{\lambda}$$



بسم الله الرحمن الرحيم

(8)

$$I = 1000 \text{ W/m}^2$$

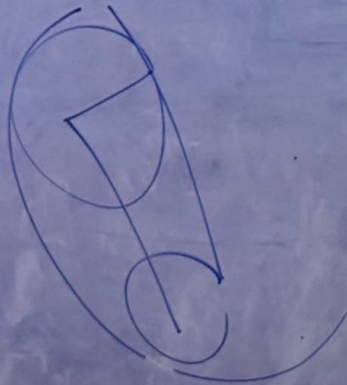
The energy per cubic meter $\rightarrow$ energy density

$$\text{energy density} = \frac{1}{2} \epsilon_0 E^2 \rightarrow \frac{1}{2} * 8.85 \times 10^{-12} * (868.3)^2 = 3.32 \times 10^{-6} \frac{\text{J}}{\text{m}^3}$$

$k \rightarrow$ wave number (m^{-1})

$$\lambda = \frac{2\pi}{k}$$

$$f = \frac{c}{\lambda}$$



$$E = \sqrt{2\mu_0 c * I} = \sqrt{240\pi * 1000} = 868.3 \text{ V/m}$$